

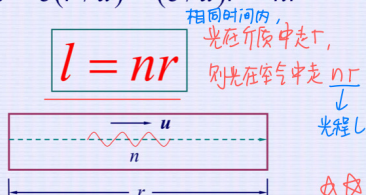


4. 光程

4.1 光程

光通过某一媒质的光程等于光在相同时间里在真空中所传播的几何路程:

$$l = c\Delta t = c(r/u) = (c/u)r = nr$$



相位差 = $2\pi \cdot \frac{\text{光程差}}{\text{真空中的波长}}$

$$\Delta\varphi = \frac{2\pi}{\lambda} (l_2 - l_1) = \begin{cases} \pm 2k\pi & \text{明} \\ \pm (2k+1)\pi & \text{暗} \end{cases}$$

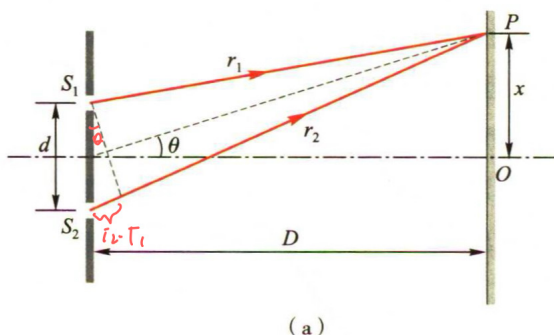
$$\delta = l_2 - l_1 = \begin{cases} \pm k\lambda & \text{明} \\ \pm (k + \frac{1}{2})\lambda & \text{暗} \end{cases}$$

由路程和成差

图的要求: $r_2 - r_1 = d \frac{x}{b}$

干涉的要求 $\delta = r_2 - r_1 = \begin{cases} k\lambda \\ (k+\frac{1}{2})\lambda \end{cases}$

$$\Rightarrow d \frac{x}{D} = \begin{cases} k\lambda \\ (k+\frac{1}{2})\lambda \end{cases}$$



$$\tan\theta = \frac{x}{b}$$

$$\delta = d \sin\theta = d \tan\theta = d \frac{x}{D}$$

$\therefore k\lambda$ 明

$(k+\frac{1}{2})\lambda$ 暗

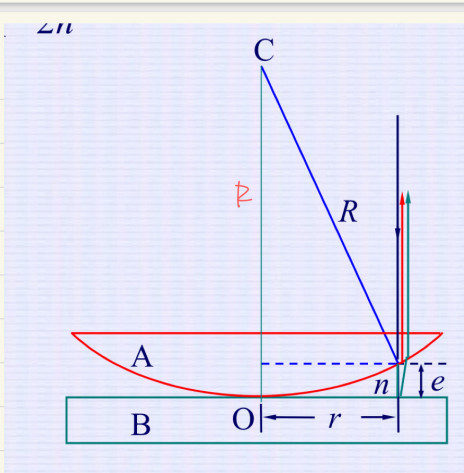
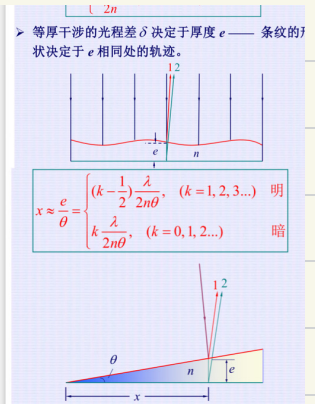
$$\cancel{\Delta x = d \sin \theta} \quad \Delta x = \frac{D \lambda}{d}$$

等厚: $2ne + \frac{\lambda}{2} = \begin{cases} k\lambda & \text{明} \\ (k+\frac{1}{2})\lambda & \text{暗} \end{cases}$

$$x = \frac{e}{\theta}$$

☆☆: $\Delta x = \frac{\Delta e}{\theta} = \frac{\lambda}{2n\theta}$

牛顿环: $2nd + \frac{\lambda}{2} = \begin{cases} k\lambda & \text{明} \\ (k+\frac{1}{2})\lambda & \text{暗} \end{cases}$



$$\begin{aligned} R^2 &= r^2 + (R-e)^2 \\ R^2 &= r^2 + R^2 - 2R \cdot e + e^2 \\ \therefore e &\ll R, r \\ \Rightarrow e &= \frac{r^2}{R} \end{aligned}$$

$$\therefore 2ne + \frac{\lambda}{2} = \begin{cases} k\lambda \\ (k+\frac{1}{2})\lambda \end{cases}$$

$$\Rightarrow r^2 = 2R \cdot (k - \frac{1}{2}) \frac{\lambda}{2n}$$

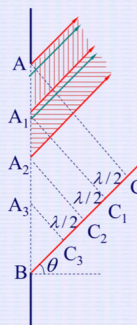
$$r \approx \sqrt{2eR} = \begin{cases} \sqrt{(k-\frac{1}{2}) \frac{R\lambda}{n}}, & (k=1, 2, 3...) \text{ 明} \\ \sqrt{\frac{kR\lambda}{n}}, & (k=0, 1, 2...) \text{ 暗} \end{cases}$$

☆☆☆: $T = \sqrt{2eR}$

2. 衍射图样

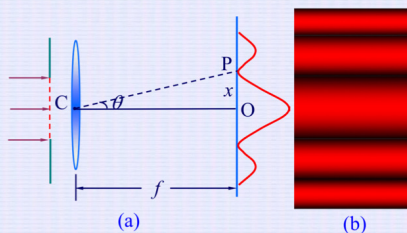
半波带法:

- 1) O: $\theta=0$ ——零级明纹中心
- 2) $BC = b \sin \theta = \pm k \lambda$ ($k=1, 2, \dots$)
—— k 级暗纹中心
- 3) $b \sin \theta = \pm (k + \frac{1}{2}) \lambda$ ($k=1, 2, \dots$)
—— k 级明纹中心
- 4) k 越大, 半波带数越多, 每个半波带面积越小。——明条纹光强随级数的增大而越小

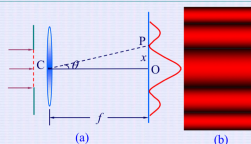


1) 明暗条件

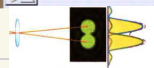
$$b \sin \theta = \begin{cases} \pm k \lambda & k \text{级暗纹} \\ 0 & \text{零级明纹} \quad (k=1, 2, 3 \dots) \\ \pm (k + 1/2) \lambda & k \text{级明纹} \end{cases}$$



$$x \approx f \sin \theta = \begin{cases} \pm k \frac{\lambda}{b} f & k \text{级暗纹} \\ 0 & \text{零级明纹} \quad (k=1, 2, 3 \dots) \\ \pm (k + \frac{1}{2}) \frac{\lambda}{b} f & k \text{级明纹} \end{cases}$$



以O为对称, 相互平行、明暗相间的条纹, 明纹亮度不一, 级数越大, 亮度越小。



点间距较大, 所形成的两像没有重叠, 看得出这是两点的像

图 12-37

张角) 为

$$\theta_0 = 1.22 \frac{\lambda}{D} \quad (12-35)$$

我们把 θ_0 称为分辨限角 (limiting angle of resolution), 也称最小分辨角; 把分辨限角的倒数称为分辨本领 (resolving power), 也称分辨率, 用 R 表示, 即

$$R = \frac{1}{\theta_0} = \frac{D}{1.22 \lambda} \quad (12-36)$$

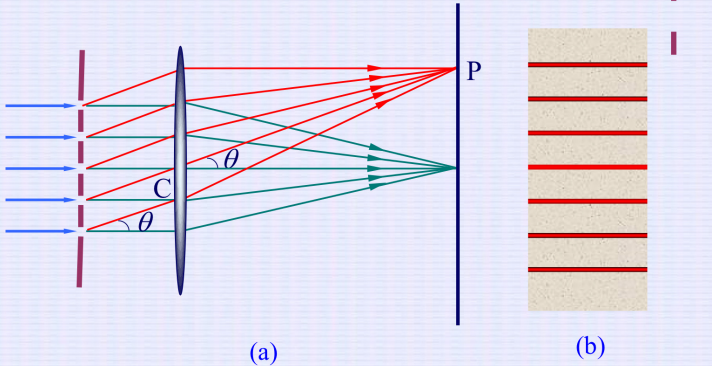


1. 条纹的形成

1.1 实验装置

光栅参数：缝宽： b 光栅常数： $d = b + b'$

缝数： N



讨论：

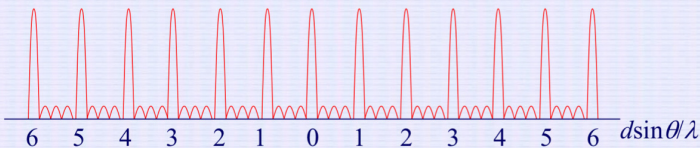
1) $\nu = \pm k\pi$ ($k = 0, 1, 2, \dots$)，即 $d \sin \theta = \pm k\lambda \rightarrow$

$A = NA_1, I = N^2 I_1$ —— k 级主极大

2) $\nu = \pm(k + \frac{m}{N})\pi$ ，即 $d \sin \theta = \pm(k + \frac{m}{N})\lambda$ ， $\begin{cases} k = 0, 1, 2, \dots \\ m = 1, 2, \dots, N-1 \end{cases} \rightarrow$

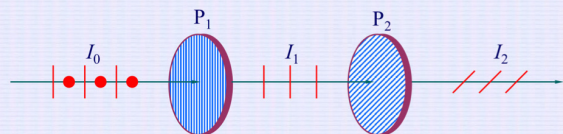
$A = 0, I = 0$ —— 两相邻主极大之间有 $N-1$ 个暗纹中心

3) 两主极大之间共有 $N-2$ 个次极大



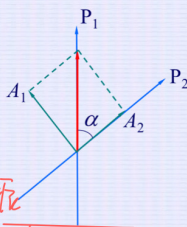
$\frac{d}{b}$ 是整数倍时，出现缺级现象

2. 马吕斯定律



$$A_2 = A_1 \cos \alpha$$

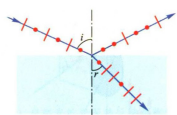
$$I_2 = I_1 \cos^2 \alpha$$



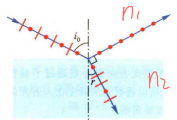
➤ $I_1 = I_0 / 2$ 自然光经过偏振片

光强变成1/2

旋转30°, 为2次最暗, 2次最亮
第4个偏振片



(a) 自然光入射于介质的表面, 入射角为 i ($i \neq 0$), 则反射光和折射光都是部分偏振光



(b) 自然光以布儒斯特角入射, 反射光为线偏振光, 这时反射光与折射光线成 90° 夹角

图 12-55

早在 19 世纪初期, 人们就在实验中发现, 当一束自然光以任意入射角 i 入射到某种介质的表面上发生反射和折射时, 其反射光和折射光一般都为部分偏振光, 其中反射光是以垂直于入射面的光振动占优, 而折射光则是以平行于入射面的光振动占优, 如图 12-55(a) 所示。

1812 年, 英国科学家布儒斯特 (S.D. Brewster, 1781—1868) 在实验中发现, 反射光的偏振化程度随着光的入射角发生变化而变化。当入射角为某一特定的角度 i_0 时, 反射光变成了线偏振光, 光振动垂直于入射面, 如图 12-55(b) 所示。这个特殊的人射角称为起偏角 (polarizing angle), 又称布儒斯特角 (Brewster's angle)。实验进一步指出, 当入射光以起偏角 i_0 入射时反射光线与折射光线正好相互垂直, 即有

$$i_0 + r = 90^\circ$$

式中的 r 为折射角。设入射光所在介质的折射率为 n_1 , 折射光所在介质的折射率为 n_2 , 由折射定律, 有

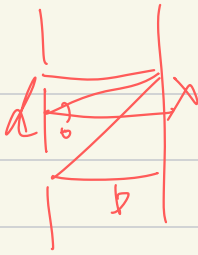
$$n_1 \sin i_0 = n_2 \sin r = n_2 \sin (90^\circ - i_0) = n_2 \cos i_0$$

$$\tan i_0 = \frac{n_2}{n_1}$$

(12-42)

上式所反映的规律称为布儒斯特定律 (Brewster's law)。

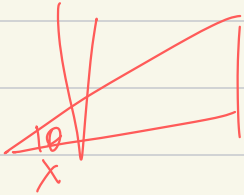
干涉?



$$\delta = l_2 - l_1 = \begin{cases} k\lambda & \text{亮} \\ (k+\frac{1}{2})\lambda & \text{暗} \end{cases}$$

$$\delta = d \sin \theta = d \tan \theta \\ \approx d \cdot \frac{x}{b}$$

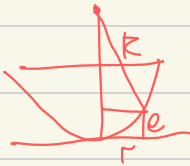
衍射?



$$\delta = 2ne + \frac{1}{2}\lambda = \begin{cases} k\lambda & \text{亮} \\ (k+\frac{1}{2})\lambda & \text{暗} \end{cases}$$

$$\theta = \frac{e}{x} \quad x = \frac{e}{\theta}$$

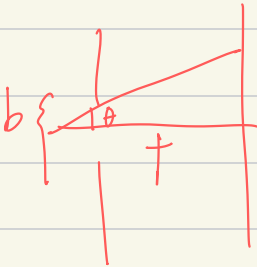
牛顿环?



$$\delta = 2ne + \frac{1}{2}\lambda = \begin{cases} k\lambda & \text{亮} \\ (k+\frac{1}{2})\lambda & \text{暗} \end{cases}$$

$$r = \sqrt{2Re}$$

衍射:

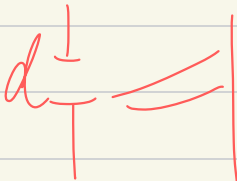


$$b \sin \theta = \begin{cases} k\lambda & \text{暗} \\ 0 & \text{亮} \\ (k+\frac{1}{2})\lambda & \text{亮} \end{cases}$$

$$\theta \approx \frac{x}{f}$$

$$x = f \sin \theta$$

光栅



$$d \sin \theta = k\lambda \quad \text{主极大}$$

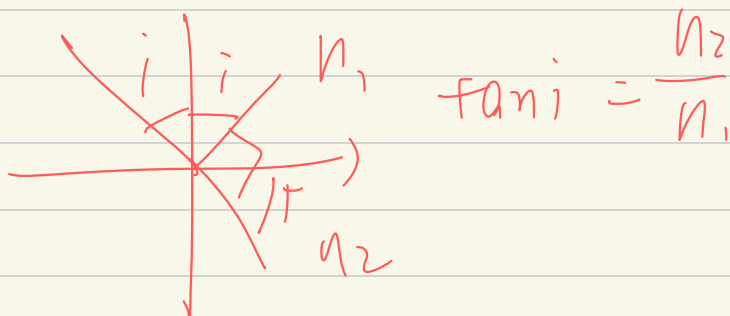
$$N-1 \quad \text{暗} \quad n-2 \quad \text{次极大}$$

$$\frac{d}{b} \quad \text{缺级}$$

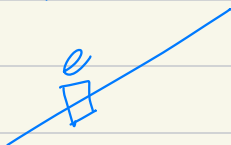
马耶斯: $I = \frac{1}{2}I$

$I = \omega \sqrt{2} I$

抑压: ~~加~~ $i + r = \int_0^v$



例 1: $L = n \tau$



$$L = \tau + (n-1)e$$

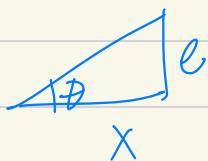
$$\delta = L \cdot \theta = \begin{cases} k\lambda & \text{亮} \\ (k + \frac{1}{2})\lambda & \text{暗} \end{cases}$$

$$\delta = d \sin \theta = d \tan \theta$$

$$= d \frac{x}{D} = \begin{cases} k\lambda \\ (k + \frac{1}{2})\lambda \end{cases}$$

例 2:

$$2ne \sin \theta = \begin{cases} k\lambda & \text{亮} \\ (k + \frac{1}{2})\lambda & \text{暗} \end{cases}$$



$\approx$

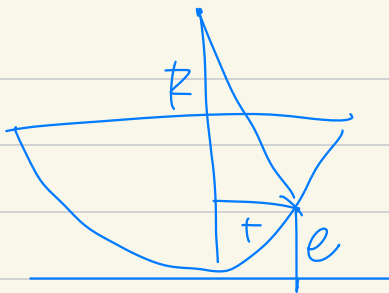
$$\tan \theta = \frac{e}{x}$$

$$\therefore x = \frac{e}{\theta}$$

$$2n \theta = \frac{k}{2n\theta}$$

$$\Delta x = \frac{\lambda}{2n\theta}$$

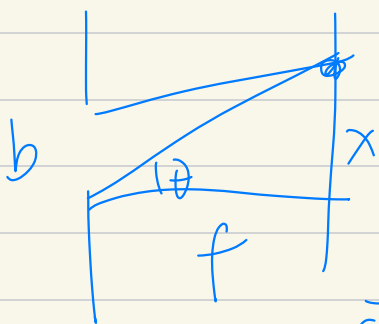
透镜环



$$t = \sqrt{2RE}$$

$$2\pi n e + \phi \frac{\lambda}{2\pi} = \begin{cases} 2\pi \frac{\lambda}{2\pi} \\ \frac{\lambda}{2\pi} \end{cases}$$

衍射:



$$b \sin \theta = \begin{cases} \frac{\lambda}{2\pi} \text{ 暗} \\ 0 \text{ 亮} \\ \frac{\lambda}{2\pi} \text{ 亮} \end{cases}$$

$$\sin \theta = \tan \theta = \frac{x}{f}$$

$$\theta = \frac{1.22 \lambda}{b}$$

$$\therefore x = f \sin \theta = \begin{cases} \frac{\lambda}{2\pi} \\ 0 \end{cases}$$

光栅

